THE COLIMIT OF HOM OUT OF A DELIGNE SYSTEM COMPUTES HOM ON THE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G2H (sources/stacks-duality/pinned-original-source.tex line 7920).

Allowed standard prerequisites (editorial).

- Deligne systems and their constant values on the open. Same chapter, the definition of a Deligne system preceding this lemma, invoked in the statement. Fixes the compatible system (K_n) and its value K.
- Exactness of filtered colimits and the long exact sequence of Hom. Derived Categories, Lemma labelled derived-lemma-representable-homological and Algebra, Lemma labelled algebra-lemma-directed-colimit-exact, cited at proof lines 7939-7943. Justifies the 5-lemma reduction along distinguished triangles.
- Truncation triangles and the identification of the bounded coherent derived category with the derived category of coherent sheaves. Derived Categories, Remark labelled derived-remark-truncation-distinguished-triangle, cited at proof lines 7944-7946, and Derived Categories of Schemes, Proposition labelled perfect-proposition-DCoh, cited at proof lines 7998-8002. Reduces to one nonzero cohomology sheaf and lets Ext be computed in the abelian category.
- Artin-Rees and the vanishing of Hom over the open. Cohomology of Schemes, Lemmas labelled coherent-lemma-Artin-Rees, coherent-lemma-homs-over-open and coherent-lemma-power-ideal-kills-sheaf, cited at proof lines 8016-8018, 7968-7970 and 8078-8080. Supplies the pro-isomorphism of the two inverse systems, the base case i = 0 and the final containment.
- Extending a coherent module from an open. Properties, Lemma labelled properties-lemma-extend-finite-presentation, cited at proof lines 8034-8038. Realises the auxiliary map over U by a map over X in the surjectivity argument.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7948 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 00DB: sources/stacks-chow/tags.txt, source lines 491–491. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7949 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 08J5: sources/stacks-chow/tags.txt, source lines 10838–10838. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7954 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01YB: sources/stacks-chow/tags.txt, source lines 2550–2550. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7985 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 0FDB: sources/stacks-chow/tags.txt, source lines 19211–19211. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7997 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01YA: sources/stacks-chow/tags.txt, source lines 2549–2549. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8016 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01PI: sources/stacks-chow/tags.txt, source lines 2231–2231. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8055 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01Y9: sources/stacks-chow/tags.txt, source lines 2548–2548. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8083 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $j : U \to X$ be an open immersion of Noetherian schemes. Let (K_n) be a Deligne system and denote $K \in D^b_{Coh}(\mathcal{O}_U)$ the value of the constant system $(K_n|_U)$. Let L be an object of $D^b_{Coh}(\mathcal{O}_X)$. Then* $\operatorname{colim} \mathrm{Hom}_X(K_n, L) = \mathrm{Hom}_U(K, L|_U)$.

Proof. Let $L \to M \to N \to L[1]$ be a distinguished triangle in $D^b_{Coh}(\mathcal{O}_X)$. Then we obtain a commutative diagram

$$\begin{array}{ccccccc} \ldots & \longrightarrow & \operatorname{colim} \operatorname{Hom}_X(K_n, L) & \longrightarrow & \operatorname{colim} \operatorname{Hom}_X(K_n, M) & \longrightarrow & \operatorname{colim} \operatorname{Hom}_X(K_n, N) \longrightarrow \ldots \\ & & \downarrow & & \downarrow & & \downarrow \\ \ldots & \longrightarrow & \operatorname{Hom}_U(K, L|_U) & \longrightarrow & \operatorname{Hom}_U(K, M|_U) & \longrightarrow & \operatorname{Hom}_U(K, N|_U) \longrightarrow \ldots \end{array}$$

whose rows are exact by Derived Categories, Lemma 0149 and Algebra, Lemma 00DB. Hence if the statement of the lemma holds for $N[-1]$, L, N, and $L[1]$ then it holds for M by the 5-lemma. Thus, using the distinguished triangles for the canonical truncations of L (see Derived Categories, Remark 08J5) we reduce to the case that L has only one nonzero cohomology sheaf.

Choose a bounded complex $\mathcal{F}^\bullet$ of coherent $\mathcal{O}_X$-modules and a quasi-coherent ideal $\mathcal{I} \subset \mathcal{O}_X$ cutting out $X \setminus U$ such that K_n is represented by $\mathcal{I}^n\mathcal{F}^\bullet$. Using "stupid" truncations we obtain compatible termwise split short exact sequences of complexes

$$0 \to \sigma_{\geq a+1}\mathcal{I}^n\mathcal{F}^\bullet \to \mathcal{I}^n\mathcal{F}^\bullet \to \sigma_{\leq a}\mathcal{I}^n\mathcal{F}^\bullet \to 0$$

which in turn correspond to compatible systems of distinguished triangles in $D^b_{Coh}(\mathcal{O}_X)$. Arguing as above we reduce to the case where $\mathcal{F}^\bullet$ has only one nonzero term. This reduces us to the case discussed in the next paragraph.

Given a coherent $\mathcal{O}_X$-module $\mathcal{F}$ and a coherent $\mathcal{O}_X$-module $\mathcal{G}$ we have to show that the canonical map

$$\operatorname{colim} \operatorname{Ext}^i_X(\mathcal{I}^n\mathcal{F}, \mathcal{G}) \longrightarrow \operatorname{Ext}^i_U(\mathcal{F}|_U, \mathcal{G}|_U)$$

is an isomorphism for all $i \geq 0$. For $i = 0$ this is Cohomology of Schemes, Lemma 01YB. Assume $i > 0$.

Injectivity. Let $\xi \in \operatorname{Ext}^i_X(\mathcal{I}^n\mathcal{F}, \mathcal{G})$ be an element whose restriction to U is zero. We have to show there exists an $m \geq n$ such that the restriction of ξ to $\mathcal{I}^m\mathcal{F} = \mathcal{I}^{m-n}\mathcal{I}^n\mathcal{F}$ is zero. After replacing $\mathcal{F}$ by $\mathcal{I}^n\mathcal{F}$ we may assume $n = 0$, i.e., we have $\xi \in \operatorname{Ext}^i_X(\mathcal{F}, \mathcal{G})$ whose restriction to U is zero. By Derived Categories of Schemes, Proposition 0FDB we have $D^b_{Coh}(\mathcal{O}_X) = D^b(Coh(\mathcal{O}_X))$. Hence we can compute the Ext group in the abelian category of coherent $\mathcal{O}_X$-modules. This implies there exists an surjection $\alpha : \mathcal{F}'' \to \mathcal{F}$ such that $\xi \circ \alpha = 0$ (this is where we use that $i > 0$). Set $\mathcal{F}' = \operatorname{Ker}(\alpha)$ so that we have a short exact sequence

$$0 \to \mathcal{F}' \to \mathcal{F}'' \to \mathcal{F} \to 0$$

It follows that ξ is the image of an element $\xi' \in \operatorname{Ext}^{i-1}_X(\mathcal{F}', \mathcal{G})$ whose restriction to U is in the image of $\operatorname{Ext}^{i-1}_U(\mathcal{F}''|_U, \mathcal{G}|_U) \to \operatorname{Ext}^{i-1}_U(\mathcal{F}'|_U, \mathcal{G}|_U)$. By Artin-Rees the inverse systems $(\mathcal{I}^n\mathcal{F}')$ and $(\mathcal{I}^n\mathcal{F}'' \cap \mathcal{F}')$ are pro-isomorphic, see Cohomology of Schemes, Lemma 01YA. Since we have the compatible system of short exact sequences

$$0 \to \mathcal{F}' \cap \mathcal{I}^n\mathcal{F}'' \to \mathcal{I}^n\mathcal{F}'' \to \mathcal{I}^n\mathcal{F} \to 0$$

we obtain a commutative diagram

$$\begin{array}{ccccc} \operatorname{colim} \operatorname{Ext}^{i-1}_X(\mathcal{I}^n\mathcal{F}'', \mathcal{G}) & \longrightarrow & \operatorname{colim} \operatorname{Ext}^{i-1}_X(\mathcal{F}' \cap \mathcal{I}^n\mathcal{F}'', \mathcal{G}) & \longrightarrow & \operatorname{colim} \operatorname{Ext}^i_X(\mathcal{I}^n\mathcal{F}, \mathcal{G}) \\ \downarrow & & \downarrow & & \downarrow \\ \operatorname{Ext}^{i-1}_U(\mathcal{F}''|_U, \mathcal{G}|_U) & \longrightarrow & \operatorname{Ext}^{i-1}_U(\mathcal{F}'|_U, \mathcal{G}|_U) & \longrightarrow & \operatorname{Ext}^{i-1}_U(\mathcal{F}|_U, \mathcal{G}|_U) \end{array}$$

with exact rows. By induction on i and the comment on inverse systems above we find that the left two vertical arrows are isomorphisms. Now ξ gives an element in the top right group which is the image of ξ' in the middle top group, which in turn

maps to an element of the bottom middle group coming from some element in the left bottom group. We conclude that ξ maps to zero in $\mathrm{Ext}^i_X(\mathcal{I}^n\mathcal{F}, \mathcal{G})$ for some n as desired.

Surjectivity. Let $\xi \in \mathrm{Ext}^i_U(\mathcal{F}|_U, \mathcal{G}|_U)$. Arguing as above using that $i > 0$ we can find an surjection $\mathcal{H} \to \mathcal{F}|_U$ of coherent $\mathcal{O}_U$-modules such that ξ maps to zero in $\mathrm{Ext}^i_U(\mathcal{H}, \mathcal{G}|_U)$. Then we can find a map $\varphi : \mathcal{F}'' \to \mathcal{F}$ of coherent $\mathcal{O}_X$-modules whose restriction to U is $\mathcal{H} \to \mathcal{F}|_U$, see Properties, Lemma 01PI. Observe that the lemma doesn't guarantee φ is surjective but this won't matter (it is possible to pick a surjective φ with a little bit of additional work). Denote $\mathcal{F}' = \mathrm{Ker}(\varphi)$. The short exact sequence

$$0 \to \mathcal{F}'|_U \to \mathcal{F}''|_U \to \mathcal{F}|_U \to 0$$

shows that ξ is the image of ξ' in $\mathrm{Ext}^{i-1}_U(\mathcal{F}'|_U, \mathcal{G}|_U)$. By induction on i we can find an n such that ξ' is the image of some ξ'_n in $\mathrm{Ext}^{i-1}_X(\mathcal{I}^n\mathcal{F}', \mathcal{G})$. By Artin-Rees we can find an $m \geq n$ such that $\mathcal{F}' \cap \mathcal{I}^m\mathcal{F}'' \subset \mathcal{I}^n\mathcal{F}'$. Using the short exact sequence

$$0 \to \mathcal{F}' \cap \mathcal{I}^m\mathcal{F}'' \to \mathcal{I}^m\mathcal{F}'' \to \mathcal{I}^m \mathrm{Im}(\varphi) \to 0$$

the image of ξ'_n in $\mathrm{Ext}^{i-1}_X(\mathcal{F}' \cap \mathcal{I}^m\mathcal{F}'', \mathcal{G})$ maps by the boundary map to an element ξ_m of $\mathrm{Ext}^i_X(\mathcal{I}^m \mathrm{Im}(\varphi), \mathcal{G})$ which maps to ξ. Since $\mathrm{Im}(\varphi)$ and $\mathcal{F}$ agree over U we see that $\mathcal{F}/\mathcal{I}^m \mathrm{Im}(\varphi)$ is supported on $X \setminus U$. Hence there exists an $l \geq m$ such that $\mathcal{I}^l\mathcal{F} \subset \mathcal{I}^m \mathrm{Im}(\varphi)$, see Cohomology of Schemes, Lemma 01Y9. Taking the image of ξ_m in $\mathrm{Ext}^i_X(\mathcal{I}^l\mathcal{F}, \mathcal{G})$ we win. $\square$